

7.12)



$$|V| = 120 \text{ V rms} \quad f = 60 \text{ Hz}$$

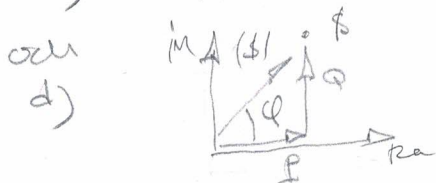
$$|I| = 12 \text{ A rms}$$

$$P = 800 \text{ W}$$

a) $P_f = \frac{P}{V_{rms} I_{rms}} = \frac{800}{120 \cdot 12} \approx \underline{0.556}$

b) $\phi ? \quad P_f = \cos \phi \Rightarrow \phi = \arccos(P_f)$
 $\phi \approx 56.25^\circ$

c) NÄTVERKETS IMPEDANS Z ?



$$\tan \phi = \frac{Q}{P}$$

$$Q = P \tan \phi = 800 \tan(56.25^\circ)$$

$$\approx \underline{1197.3 \text{ VAR}}$$

or $S = V_{rms} I_{rms}^* = |V_{rms}|^2 \frac{1}{Z^*}$

or $S = P + jQ$

$$\Rightarrow P + jQ = |V_{rms}|^2 \frac{1}{Z^*}$$

$$Z^* = \frac{|V_{rms}|^2}{P + jQ}$$

ELER $Z = \frac{|V_{rms}|^2}{P - jQ} = |V_{rms}|^2 \frac{P + jQ}{P^2 + Q^2}$

$$Z = 120^2 \frac{800 + j1197.3}{800^2 + 1197.3^2} \approx \underline{5.56 + j8.31 \Omega}$$

SVAR: a) $P_f = 0.556$ c) $Z = 5.56 + j8.31 \Omega$

b) $\phi = 56.25^\circ$ d) $R = 5.56 \Omega$

$X = 8.31 \Omega$